SYSC 3600 - Systems and Simulation

Properties of linear systems. Linear dynamic models of engineering systems. Applications of the Laplace transform. Transfer functions. Block diagrams. Frequency and time response. System simulation with digital computers.

SYSC 4005 - Discrete Simulation/Modelling

Simulation as a problem solving tool. Random variable generation, general discrete simulation procedure: event table and statistical gathering. Analyses of simulation data: point and interval estimation. Confidence intervals. Overview of modeling, simulation, and problem solving using SIMSCRIPT, MODSIM, and other languages.

SYSC 4102 - Performance Engineering

Same PROF as Requirements Engineering. Course content probably boring and not directly applicable. Techniques based on measurements and models, for predicting and evaluating the performance of computer systems. Instrumentation. Simple queueing models and approximations. Techniques for modifying software designs to improve performance.

SYSC 4111 - Formal Methods in Software Engineering

Introduction to formal methods in software engineering with coverage of propositional and first-order logic (syntax, semantics, proof theory), formal specification languages, bounded analysis and validation, formal specification tools, and model checking with finite-state machines, temporal logic, and model checking tools.

SYSC 4130 - Human Computer Interaction

User-centric design, evaluation, and implementation of interactive computing systems. Topics include: designing, prototyping, implementing, and evaluating user-facing systems and interfaces; data gathering, analysis, and interpretation; persuasive design; dark patterns; accessibility; design for security and privacy.

SYSC 4205 - Image Processing for Medical Applications

Students should have knowledge of Fourier series. Requires restudying MATH 1005 Two-dimensional signals, filters, and Fourier transforms. Image acquisition, sampling, quantization and representation. Image perception. Digital and film cameras. Medical imaging technologies. Image processing operations: histogram, convolution, morphological, segmentation, registration. Image compression and formats.

SYSC 4206 - Surgical Robotics

PREREQ: **SYSC 3600** LOOK UP SYLLABUS Surgical robotic system architecture, forward and inverse kinematics of articulated robot arms, force and position control, unilateral and bilateral teleoperation of surgical robots, haptics and force feedback, instrumentation, image-guided surgery, design and implementation of robotic systems for minimally invasive surgery.

SYSC 4502 - Communications Software

PREREQ - **SYSC 4602 (Which if a fall course only)** Communications software architectures, protocols and operating systems. Application layer protocols, APIs and socket programming. P2P algorithms, network virtualization, SDN. Reliable data transfer algorithms, FSM, MSC. Network security. Multimedia applications, RTSP, CDN, DASH, RTP, RTCP. Packet scheduling algorithms, DiffServ, IntServ, RSVP. Traffic classification, cross-layer optimization.

SYSC 4504 - Fundamentals of Web Development

WWW architecture, web servers and browsers, core protocols. Web pages, their structure, interpretation and internal representation. Client-side and server-side programming. Data representation. Interfacing with databases and other server-side services. Cookies, state management, and privacy issues. Security. Web services.

SYSC 4505 - Automatic Control Systems I

PREREQ: MATH 2004 & SYSC 3600

SYSC 4600 - Digital Communications

PREREQ: SYSC 3501 which has PREREQs MATH 3705 and SYSC 3600 and STAT 3502

SYSC 4602 - Computer Communications

FALL COURSE ONLY

SYSC 4607 - Wireless Communications

PREREQ: SYSC 3501 which has PREREQs MATH 3705 and SYSC 3600 and STAT 3502

SYSC 4906 - Special Topics - Advanced C++ Programming

Abstract C++ machine, compiling and linking; smart pointers, lvalue, rvalue, universal references, containers, ownership, semantics; generic programming, metaprogramming, concept specification; concurrency; C++ idioms; exception safety, debugging, performance; API and ABI.

SYSC 4906 - Special Topics - Artificial Intelligence in Eng

Students should: Have strong math skills, including working with differential equations, matrix operations, and gradients

This course will follow "The One Hundred Page Machine Learning Book" from cover-to- cover. Students will gain an introductory-level understanding of both supervised and unsupervised machine learning (ML), including deeper knowledge of a number of algorithms of each type. Students will learn how to evaluate and quantify predictive performance of ML systems. Students will also become familiar with one or more ML development environments with practical assignments and demonstrations.

Got info for special topics from hiring page: https://carleton.ca/sce/positions-available/

COMP elective list for winter 2025

COMP 2108 - Applied Cryptography and Authentication

Requires registration override request

COMP 3002 - Compiler Construction

The structure, organization and design of the phases of a compiler are considered: lexical translators, syntactical translators, scope handlers, type checkers, code generators and optimizers. Components of a compiler will be implemented.

COMP 3008 - Human-Computer Interaction

Requires registration override request Fundamentals of the underlying theories, design principles, development and evaluation practices of human-computer interaction (HCI). Topics may include: theories of interaction, user interface frameworks, desktop, web, mobile, and immersive applications, usability inspection and testing methods, and qualitative and quantitative approaches to HCI research.

COMP 4102 - Computer Vision

Priority to 4th year BCS students. The basic ideas and techniques of computer vision. The central theme is reconstructing 3D models from 2D images. Topics include: image formation, image feature extraction, camera models, camera calibration, structure from motion, stereo, recognition, augmented reality, image searching.

COMP 4107 - Neural Networks

Requires registration override request **PREREQ: SYSC 4415 (Introduction to Machine Learning)** An introduction to neural networks and deep learning. Theory and application of Neural Networks to problems in machine learning. Various network architectures will be discussed. Methods for improving optimization and generalization of neural networks. Neural networks for unsupervised learning

COMP 4202 - Computational Aspects of Geographic Information Systems

**PREREQ: COMP 3804 or MATH 3804

COMP 4900 E - Special Topics - Real-Time Operating Systems

This course covers the basics of Real-Time Operating Systems (RTOS). The course focuses on the primary principles of RTOS. Topics include RTOS architecture, inter process communication, determinism, real-time scheduling, interrupt latency, fast context switching, time and space partitioning in hard real-time environments.